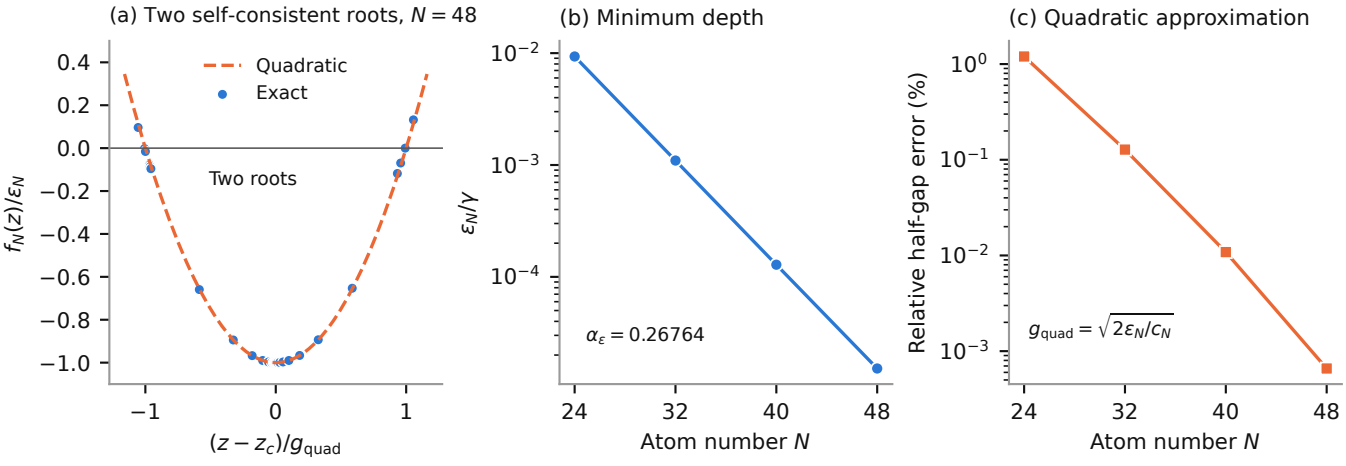
(a) Two self-consistent roots, $N = 48$
Quadratic
Exact
Two roots
$f_N(z)/\varepsilon_N$
$(z - z_c)/g_{\mathrm{quad}}$
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1
0
1
(b) Minimum depth
ε_N/γ
10^{-2}
10^{-3}
10^{-4}
$\alpha_\varepsilon = 0.26764$
24
32
40
48
Atom number N
(c) Quadratic approximation
Relative half-gap error (%)
10^{0}
10^{-1}
10^{-2}
10^{-3}
$g_{\mathrm{quad}} = \sqrt{2\varepsilon_N/c_N}$
24
32
40
48
Atom number N